



HABIB UNIVERSITY
Digital Logic and Design
EE-172 / CS-130 / CE-222
Fall 2025 (L2)

Quiz # 2

Student ID: _____

Student Name: _____

Date: October 16, 2025

Time: 25 minutes

Max points: 16

Instructions:

- Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- Answer sheets should contain all steps, working, explanations, and assumptions.**
- Attempt the quiz in the spaces provided for each question.
- You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

- | No. | Questions | Pts |
|-----|---|-----|
| 1 | Select the best option which represents the standard SOP expression for $F(A,B,C) = \pi(0,1,7) + d(4)$. | [1] |
| | <ul style="list-style-type: none">▪ $\bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + A \cdot B \cdot C$▪ $\bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot C + A \cdot B \cdot C$▪ $\bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + A \cdot \bar{B} \cdot C + A \cdot B \cdot \bar{C}$▪ $\bar{A} \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C + \bar{A} \cdot B \cdot C + A \cdot \bar{B} \cdot C + A \cdot B \cdot \bar{C}$ | |
| 2 | Simplify using k-map: $F(A,B,C,D) = \pi(0,1,2,6,8,9,14,15) + d(7)$ | [4] |

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	1	0
$\bar{A}B$	1	1	X	0
AB	1	1	0	0
$A\bar{B}$	0	0	1	1

$$B\bar{C} + A\bar{B}C + \bar{A}CD$$

Or

$$B\bar{C} + A\bar{B}C + \bar{B}CD$$

- 3 A water-well pump is to be turned on automatically if the water level is low in any two or more of the three reservoirs it supplies. Water-level detectors in the reservoirs each generate a logic 0 when the level in the reservoir is low.
- Design a logic system that turns on the pump by generating a logic 1 in response to inputs received from the level detectors. Use Boolean theorems to obtain a simplified expression, showing all steps clearly. Also, draw the logic diagram for the simplified expression. [8]
 - Implement the simplified expression using NAND gates only. Then, briefly comment (in not more than three lines) on whether the NAND implementation is beneficial in this case, considering factors such as gate count, circuit complexity, and space efficiency. [3]

[Use backside of this sheet to answer this question.]

Let A , B and C be the Boolean variables representing the water level in the three reservoirs and F be the output.

Part a

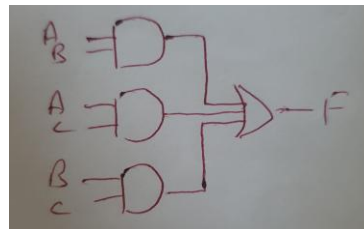
Truth Table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

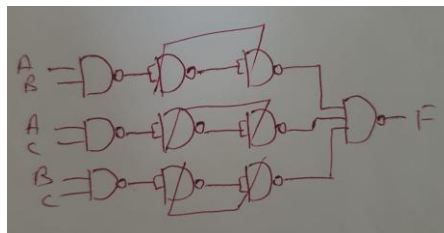
$$\begin{aligned}
 F(A, B, C) &= \sum (3, 5, 6, 7) \\
 &= \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC \\
 &= \bar{A}BC + A\bar{B}C + AB(\bar{C} + C) \\
 &= \bar{A}BC + A\bar{B}C + AB(1) \\
 &= \bar{A}BC + A\bar{B}C + AB \\
 &= \bar{A}BC + A(\bar{B}C + B) \\
 &= \bar{A}BC + A(C + B) \\
 &= \bar{A}BC + AC + AB \\
 &= C(\bar{A}B + A) + AB \\
 &= C(B + A) + AB \\
 &= BC + AC + AB
 \end{aligned}$$

Part b

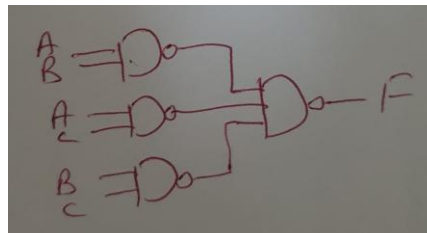
Mixed gate implementation:



NAND equivalent:



Simplified NAND equivalent:



Both the mixed-gate and NAND-equivalent circuits use four gates. However, NAND gates occupy less chip area; therefore, the NAND-based circuit is more efficient in terms of space and power consumption.